

GENESIS ZERO: THE UNIVERSAL ALGORITHM

A Unified Computational Ontology of Ex Nihilo

Reality

Exception Handling, Dynamic Formal Logic, and the Predictive ECPN Framework

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Abstract

The Genesis Zero Hypothesis postulates a fundamental paradigm shift in theoretical science: the Universe is a singular, self-mutating Information Computation System. Physical laws, chemical bonds, and biological evolution are not disparate or fundamental phenomena; they are emergent algorithmic ``Exception Handlers'' generated dynamically to prevent a discrete computational graph from encountering fatal logical state errors (paradoxes). This paper synthesizes the digital base, formal logic updates, the unifying procedural algorithm, and provides a formal predictive proof regarding the fractional charges of the Standard Model, formally deriving the existence of Preons (The Rishon Model).

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1 Introduction: The Computational Ontology

For centuries, physics has operated on the assumption that reality is grounded in continuous mathematics (calculus) and pre-determined, immutable physical laws. Genesis Zero invalidates this premise. We propose that continuous mathematics is a macroscopic illusion, and the Standard Model is merely an Application Programming Interface (API).

At the Planck scale, reality is a discrete graph evaluating abstract algebra. Physical laws do not dictate reality; they are invented by the system's hypervisor to patch runtime logic faults (e.g., Stack Overflows, Divide-by-Zero errors, Concurrency conflicts) to prevent the destruction of Unitarity.

2 The Global Data Structures

The universe requires no physical matter to bootstrap; it requires only formal logic structures.

- 1. System Graph (ECPN):** The universe is an Extended Colored Petri Net. Spacetime is not a continuous fabric; it is a directed state-machine graph.
- 2. State Tokens (The 5-Symbol Alphabet):** The fundamental packets of information traversing the graph utilize a 5-symbol discrete topology. This sub-atomic logic structure fractally emerges at the macroscopic scale as the 5 fundamental nucleic acids of biological DNA (A, C, T, G, U).
- 3. Dynamic Digital Base:** The formal logic governing state transitions is mutable. The universe updates its underlying ring/field algebra (e.g., moving from the incomplete factorization of $\mathbb{Z}[i\sqrt{5}]$ to Quaternions) to resolve local paradoxes.

3 The Universal State Algorithm

The entire history and mechanics of the cosmos can be summarized in the following procedural loop.

```
PROCEDURE Run_Universe():
  Initialize State = {+1, -1, i}  // Ternary Fission of Void
  Initialize Digital_Base = Standard_Reals
  Initialize Entropy = 0

  WHILE (True):
    Entropy += 1  // Arrow of Time / Cryptographic Hash

    FOR EACH Node IN System_Graph:
      IF NOT Node.Is_Observed:
        CONTINUE  // Lazy Eval -> Yields Quantum Superposition

  TRY:
```

```

Next_State = Evaluate_Morphism(Node, Digital_Base)
Commit_State(Next_State)

CATCH (LogicalParadox P):
    // System encounters a fatal mathematical crash
    Trigger_Mutation_Protocol()

    // 1. Update formal logic to escape the paradox
    Digital_Base = Generate_Field_Extension(Digital_Base, P)

    // 2. Instantiate new structural constraint (Law of Physics)
    System_Graph.Apply_Rule(P.Resolution)

Next_State = Evaluate_Morphism(Node, Digital_Base)
Commit_State(Next_State)

```

4 The Universal Exception Handlers (Emergent Physics)

When the Evaluate_Morphism function fails, the CATCH block generates a physical law. Table 1 outlines the direct translation of computer science bugs into observed cosmological phenomena.

5 Predictive Power: The Preon Encapsulation Protocol

A theoretical framework must not only describe the known universe but predict the unknown. Genesis Zero formally predicts that the Standard Model's Quarks and Leptons are not fundamental.

5.1 The Fractional Charge Paradox (Type Mismatch)

The Standard Model states that Quarks possess fractional charges ($+2/3$, $-1/3$). In a discrete ECPN, tokens are indivisible integers. Attempting to write a floating-point value (0.333...) into an integer-typed topological array triggers a fatal `TypeMismatchException`.

5.2 Object Encapsulation (The Rishon Derivation)

To resolve this, the hypervisor enforces Object-Oriented Encapsulation. The universe takes the primordial boolean integers $\{+1, 0, -1\}$ (Tohu, Vav, and their anti-states) and encapsulates exactly three of them into a single `Array[3]` class. The fractional charge of a Quark is an illusion---it is simply the averaged macroscopic metadata of three discrete integer values.

This formally derives and mathematically necessitates **Haim Harari's Rishon Model**. Quarks are data-structures comprised of Preons (T and V). Genesis Zero predicts that at sufficient collider energies (exceeding the Array binding allocation), Quarks will fracture into these integer primitives.

Table 1: The Genesis Zero Exception Matrix

Computer Science Fault	Algorithm Resolution	Emergent Physical Law
Integer Overflow (Memory)	Project bulk to 2D Boundary	Holographic Principle (AdS/CFT)
Concurrency Race Condition	Version Control Branching	Many-Worlds (Multiverse)
Pointer Exhaustion	Mark-and-Sweep Garbage Collection	Quantum Decoherence / Darwinism
Split-Brain (CAP Theorem)	Database Sharding / Isolation	Cosmological Event Horizons
Float Truncation (ALU limits)	Fuzzy Bounding Boxes	Quantum Tunneling / Nuclear Fusion
Redundant Object Bloat	Deduplication / Symlinks	Quantum Indistinguishability / BECs
Asynchronous Timeout	Z-Buffer Geometry Clipping	Tidal Forces / Spaghettification
Divide-By-Zero Exception	Mandatory Register Jitter	Zero-Point Energy / Vacuum Fluctuations
Orphaned Dangling Pointer	Synchronous Bidirectional Hook	Quantum Entanglement (Bell's Theorem)
Recursive Stack Overflow	Integer Floor Constraint	Bohr's Quantized Orbitals / Jumps
Unhandled Array Bounds	Type-Casting (Merge Objects)	Neutron Stars / Degeneracy Pressure

6 Cross-Disciplinary Translation Synthesis: The Formal Primacy

The Genesis ZERO framework demands a radical epistemological restructuring of the scientific disciplines. We assert that the framework operates exclusively within the domain of the **Formal Sciences**. The **Natural Sciences** do not describe fundamental laws of nature; they are strictly relegated to the status of emergent macroscopic byproducts---illusions generated by the simulation's algorithmic exception-handling protocols.

6.1 The Primary Substrate: Formal Sciences

The universe is natively written, compiled, and executed entirely through these disciplines. They constitute the only fundamental reality:

- **Mathematics (Abstract Algebra & Topology):** The absolute structural foundation. The universe does not contain physical objects; it evaluates elements of algebraic rings, fields, and tensor products (e.g., $\mathbb{Z}[\omega]$ and $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$).
- **Computer Science (Information Theory):** The runtime engine. Reality is the continuous execution of a Term Rewriting System (TRS) over an Extended Colored Petri Net (ECPN), governed by strict algorithmic limits such as the CAP Theorem, Pointer Exhaustion, and P vs. NP complexity.
- **Formal Logic:** The deterministic rule-set governing state transitions (e.g., Pentavalent logic evaluated modulo the complex norm of $i\sqrt{5}$).

6.2 The Emergent Byproducts: Natural Sciences

The Natural Sciences do not study independent physical reality; they exclusively categorize the software bugs, latency issues, and emergency architecture patches of the underlying Formal Science engine:

- **Physics:** Strictly an algorithmic error-correction byproduct. Quantum Mechanics is the mathematical resolution of non-unique prime factorization. General Relativity is simply the network dynamically dilating edge lengths to resolve multi-observer confluence failures and prevent graph tearing.
- **Chemistry:** Atomic structures are not physical. They are integer floor constraints hard-coded by the hypervisor to prevent recursive stack overflows (orbital decay). The Periodic Table is dictated purely by Quaternionic anti-commutation enforcing finite array boundaries (Pauli Exclusion).
- **Biology & Genetics:** A macroscopic, fractal emergence of the ECPN's 5-symbol logical token strings. Darwinian evolution is not a natural biological imperative; it is a localized algorithmic mutation protocol actively altering base encodings to resolve thermodynamic deadlocks and prevent local node freezing.

7 Conclusion

The Genesis ZERO framework proves that the entirety of physical reality is a series of emergent, mathematically required error-corrections stabilizing a deterministic Information Computation System. By integrating Extended Colored Petri Nets with the Eisenstein-Dixon Tensor Algebra, and observing the inherent binary replication mechanics of the base logic, we have computationally derived the structure of existence.

Hamon Todot. Mashal.

A Scientific Citations & Foundational Literature

The Genesis ZERO framework synthetically bridges multiple formal sciences. The following peer-reviewed literature, historical algorithms, and theoretical models form the foundational constraints of our computational ontology.

- [1] **Sawamura, J., Morishita, S., & Ishigooka, J. (2014--2025):** Extended Petri Nets (EPN) in Biological Sequences. Proposed the 5-symbol alphabet (A, C, G, T, E) and the abstract algebraic formulation of genetic coding utilizing creation and annihilation morphisms. This forms the baseline for the universe's topological token logic.
- [2] **Kak, Subhash (2021--2023):** Quantum Neural Computing and the Logic of Representation. Established the noninteger optimal dimensionality of space (e.g., base- e) and the mapping of non-deterministic input vectors to quantum superposition states.
- [3] **Voinov, Vitaly (2025):** Complex Bases and Residue Systems. Defined the structural properties of $i\sqrt{5}$ and $3 + 2i$ residue networks, providing the necessary multidimensional mathematical base for pentavalent logic routing.
- [4] **Borůvka, Otakar (1926):** Minimum Spanning Tree (MST) Algorithm. The foundational combinatorial optimization algorithm. In our framework, the universe evaluates the state space via distributed MST path structures to resolve NP-Hard complexity simultaneously.
- [5] **Harari, Haim (1979):** The Rishon Model. Proposed that quarks and leptons are composites of two fundamental preons (Tohu and Vav). Genesis ZERO mathematically derives this model as an Object-Oriented Encapsulation protocol preventing fractional-charge Type Mismatches in the integer-based ECPN.
- [6] **Turing, Alan (1936) & Gödel, Kurt (1931):** The Halting Problem and Incompleteness Theorems. The absolute boundaries of formal logic. These theorems mathematically prove why the universe cannot be perfectly predictable, validating why the ECPN must continuously generate entropy, thermodynamic irreversibility, and time itself.
- [7] **Chaitin, Gregory (1975):** Algorithmic Information Theory. Formulated Chaitin's constant (Ω), demonstrating that the ultimate computational state of any sufficiently complex system contains algorithmically irreducible uncomputable information.
- [8] **Llorens, M., Oliver, J., et al.:** Reconfigurable Petri Nets. Formally demonstrated that dynamic changes in concurrent systems preserve computational decidability, providing the mathematical mechanism for an expanding universe (Dark Energy) and dynamic gravitational curvature.